ChatGPT

HTNLA Recast: Rigorous Nonlinear Transdimensional Maps over a Labeled Block Substrate

Executive summary

Abstract. The topic was nominally left unspecified, but the uploaded materials make the intended research target clear: a rigorous rewrite of **Hypervector Nonlinear Transdimensional Algebra (HTNLA)** as the Nexus v4 S9 subsystem-math source. The first attachment supplies the legacy HTNLA draft; the second supplies the governing rewrite brief, canonical equations, claim-status vocabulary, compatibility requirements, and identified defects. I therefore treat the latter as the controlling specification and the former as historical source material to be audited rather than as evidence. filecite turn0file0 filecite turn0file1

The central conclusion is that **HTNLA is technically defensible, but not as a "graded nonlinear algebra."** Once arbitrary nonlinear activation is introduced, the resulting parameterized maps are not closed under operator addition or scalar multiplication in the manner required for an algebra of linear operators. The clean formulation is instead a **family of label-preserving nonlinear maps, or equivalently finite networks of labeled nonlinear morphisms, built over HTLA's block-linear substrate**. This preserves the useful idea—different source labels may feed different target labels through dimensionally typed blocks and target-specific nonlinearities—without making a false algebraic claim. The surrounding machine-learning literature shows that type-specific transformations, heterogeneous representations, sparse routing, and invertible neural architectures are already established ideas; HTNLA's defensible contribution lies in their **specific integration with HTLA's labeled transdimensional interface, explicit reconstruction epistemics, and the Nexus quinary result-status contract**, not in nonlinear activation itself. R-GCN and HGT are especially close structural precedents. [1]

The canonical layer should remain:

$$z_t = b_t + \sum_{s:(t,s)\in E} W_{t,s} x_s + r_t, \qquad y_t = \phi_t(z_t),$$

with optional target-wise sparsification

$$y_t \leftarrow \mathrm{kWTA}_t(y_t),$$

and multilayer form

$$H^{(k+1)} = \sigma^{(k)}\left(M^{(k)} H^{(k)} + b^{(k)}\right),$$

optionally augmented, where compatible label geometry exists, by

$$H^{(k+1)} = R^{(k)} H^{(k)} + \sigma^{(k)}\left(M^{(k)} H^{(k)} + b^{(k)}\right).$$

These are the Nexus v4 canonical forms supplied in the governing brief. 【filecite】turn0file1【】

The most important mathematical correction concerns **invertibility**. It should never be represented as a binary property inherited automatically from HTLA. Exact inversion survives only when the relevant composite map is injective—typically requiring dimension compatibility, an injective affine precursor on the relevant domain, injective nonlinearities, and no information-destroying sparsifier. RealNVP and related invertible-network work demonstrate that invertibility must be an architectural property engineered into the network, while SurVAE Flows is particularly relevant to HTNLA because it explicitly distinguishes bijective from surjective/non-invertible transformations and treats reverse inference accordingly. [2]

The Nexus five-way reconstruction taxonomy—**exact inverse, pseudoinverse, partial unbinding, iterative/resonator reconstruction, and explicitly unknown/ambiguous output**—should be retained, but understood as a taxonomy of reconstruction methods or guarantees. Its externally visible result must use the canonical quinary statuses `EXACT / AMBIGUOUS / UNKNOWN / IMPOSSIBLE`. This distinction is one of the strongest parts of the proposed subsystem: a reconstruction algorithm is not permitted to turn information loss into apparent certainty. 【filecite】turn0file1【】

The legacy claims that a fixed percentage of k-WTA sparsity turns dense $O(D^2)$ computation into $O(D \log D)$, or that nonlinear HTNLA has already yielded a fixed percentage improvement in associative recall, do not survive review. A post-activation top-k operation cannot retroactively remove the cost of the dense matrix multiplication that produced its input. Fixed fractional sparsity can yield a constant-factor reduction for later sparse-aware operators; an asymptotic change requires actual structural sparsity, bounded-degree label coupling, low-rank structure, FFT-compatible operators, conditional expert routing, or another mechanism whose active computation grows subquadratically. Modern sparse-network theory supports the value of selectively activating subnetworks, but not the legacy complexity derivation. [3]

Likewise, available winner-take-all literature supports **sparsification and specialized representational properties**, not a general theorem that arbitrary k-WTA after arbitrary learned nonlinear blocks "preserves concentration of measure." Winner-Take-All Autoencoders directly impose sparse representations; Dasgupta and Tosh prove useful approximation properties for a specific expand-and-sparsify construction, while also showing that ordinary WTA does **not** automatically obtain all of the manifold-adaptivity properties one might hope for. A general HTNLA concentration theorem therefore remains a `RESEARCH HYPOTHESIS`, not an established guarantee. [4]

Finally, biologically named learning mechanisms and quantum extensions should not be part of the mathematical core. Reward-modulated STDP provides legitimate literature for a **Level-2 functional analogy** in which a delayed global modulator gates locally accumulated eligibility, but the legacy HTNLA draft's particular dopamine sigmoids, risk formulas, and constants are not derived from those papers. Izhikevich, Frémaux–Sprekeler–Gerstner, and Legenstein–Pecevski–Maass support the broad mechanism, not the draft's equations. [5] Quantum nonlinear processing is even more clearly a research boundary: QSVT operates on singular values of linear operators or matrix encodings; it does not imply that an arbitrary classical nonlinear function can simply be "block-encoded as a unitary." Recent work on nonlinear quantum-state functions and neural activations instead uses multi-copy, measurement, ancilla, or hybrid quantum-classical mechanisms. [6]

Recommended disposition: ratify the labeled nonlinear-map formulation; make typed reconstruction status part of the API contract; retain k-WTA only as an optional sparsification operator; implement and benchmark before attaching performance numbers; maintain an optional explicitly invertible profile; and require every biological, energy, quantum, or performance claim to carry one of the existing Nexus claim-status terms at the confidence it has actually earned.

Scope, assumptions, and methodology

The analysis treats the uploaded rewrite specification as the source of **Nexus-internal canonical requirements**, not as independent scientific evidence. It treats the legacy HTNLA document as a historical design artifact that contains both useful mathematical structure and assertions requiring correction. The legacy document presents per-label nonlinear operators, residual connections, k-WTA, retrieval/indexing proposals, code, and benchmark targets, but also repeatedly treats internal projections as established evidence. filecite turn0file0 The companion specification explicitly requires correcting those defects and supplies the Nexus-compatible layer equations, reconstruction taxonomy, quinary binding relationship, and twelve-term status vocabulary. filecite turn0file1

The external literature search prioritized original papers and official proceedings or publisher versions. Five areas were investigated because they bound the actual novelty and correctness of HTNLA:

Research question	Primary literature emphasized	What it determines
Are type/label-specific nonlinear maps novel?	R-GCN, Heterogeneous Graph Transformer, sparse/MoE literature	Novelty boundary
How should information loss and invertibility be modeled?	RealNVP, SurVAE/invertible-network literature	Reconstruction semantics
What does k-WTA genuinely establish?	Winner-Take-All Autoencoders, expand-and-sparsify theory, sparse-network theory	Sparsity claims
Can biologically named learning rules be grounded?	Reward-modulated STDP papers	Scope of functional analogy
Is arbitrary quantum nonlinear block encoding valid?	QSVT and recent nonlinear quantum-processing work	Quantum boundary

The research also emphasized developments within approximately the five years preceding September 1, 2026. Recent work is useful not because it retroactively validates HTNLA, but because it shows where adjacent fields have moved: sparse conditional computation has acquired more formal analysis; MoE systems increasingly distinguish **parameter count from active computation and communication**; heterogeneous models continue to employ type-dependent operators; and quantum work treats nonlinearity as a nontrivial measurement/hybrid-resource problem rather than as a free unitary primitive. [7]

A second methodological rule is that mathematical claims must be one of three things: directly derivable in the paper, supported by a cited theorem under stated hypotheses, or explicitly classified as `INFERRED` or `RESEARCH HYPOTHESIS`. In particular, an empirical performance target is not evidence of achieved

performance; an architectural metaphor is not a physical law; and an internally repeated number does not become externally verified through repetition. This follows the epistemic discipline specified in the Nexus rewrite brief. filecite turn0file1

The reference implementation below was executed rather than merely inspected. Its test fixture deliberately uses **different dimensions for different labels**—the exact condition under which one of the legacy prototype functions failed. A malformed block was also tested and correctly rejected by the dimensional contract. The legacy draft and governing brief document the earlier heterogeneous-dimension and risk-modulation problems that motivated this requirement. filecite turn0file0 filecite turn0file1

Relationship to prior art and recent developments

HTNLA should make a **narrow novelty claim**. The raw architectural idea

$$\text{typed inputs} \to \text{type-dependent linear transformations} \to \text{nonlinearities} \to \text{typed outputs}$$

already has close relatives.

Relational Graph Convolutional Networks associate different transformations with different relation types in multi-relational graphs. Heterogeneous Graph Transformer goes further with node- and edge-type-dependent parameters and heterogeneous attention. These are not equivalent to HTNLA—the graph semantics and message-passing assumptions differ—but they demonstrate that **structurally typed transformations are established prior art**, so “a different nonlinearity for each label” is not itself a sufficient novelty claim. [1]

Sparse mixture-of-experts architectures provide a second neighboring family. Their important analogy is not that an “expert” equals an HTNLA label, but that a large parameter space need not imply every parameter is active for every input. The recent theoretical treatment by Baykal and colleagues formalizes data-dependent sparse subnetworks and establishes approximation results for a specific LSH-based construction. LSH-MoE, published at NeurIPS 2024, illustrates how modern MoE research separately optimizes routing and inter-device communication rather than merely equating “sparse” with asymptotically cheap. [8]

Invertible neural networks are a third and especially important neighbor. RealNVP constructs explicitly invertible, learnable transformations with exact forward and inverse computation. SurVAE Flows enlarges the design space to include surjective transformations for which one direction is deterministic while the reverse direction becomes stochastic or otherwise non-unique. Those architectures supply a much firmer conceptual basis for HTNLA's inverse taxonomy than a generic recommendation to “use gradient ascent.” [2]

Winner-take-all mechanisms are also long-established. Makhzani and Frey use WTA to enforce sparse hidden representations directly. Dasgupta and Tosh analyze an expand-and-sparsify mapping in which a high-dimensional random expansion is followed by top-k retention; they obtain approximation results under explicit conditions and, importantly for HTNLA, show that ordinary WTA does not automatically yield all desired manifold-adaptive behavior. These results support treating k-WTA as a useful operator whose effects must be established for the chosen setting—not as a generic concentration or stability theorem. [4]

Universal approximation needs similar care. Cybenko's classic result concerns sufficiently expressive feed-forward networks with continuous sigmoidal nonlinearities and enough hidden units; it is an approximation theorem for a **class of networks under hypotheses**, not a certificate that an arbitrary fixed-depth, fixed-width, block-constrained, k-sparse HTNLA instance is a universal approximator. [9] HTNLA can legitimately say that its architecture family can contain ordinary feed-forward special cases and may therefore inherit relevant approximation theorems when their conditions are actually satisfied. It should not put "Universal Approximator" in a comparison table as an unconditional property of every configured network.

A useful way to state the resulting novelty is:

> `INFERRED` **novelty claim:** HTNLA's defensible contribution is not nonlinear activation, heterogeneous blocks, sparse routing, or invertibility analysis individually. It is their combination inside the Nexus labeled-transdimensional interface, together with an explicit reconstruction-method taxonomy, a four-state epistemic output contract, and compatibility with the deterministic quinary reference operation.

That is narrower than the legacy framing but technically stronger because it identifies a combination that can be specified, implemented, and tested.

Recent developments relevant to the subsystem

Year	Development	Relevance to HTNLA
2022	Baykal et al. formalized a model of data-dependent sparsely activated networks and proved approximation results for an LSH-routed construction. [3]	Supports rigorous sparse conditional computation, but only under explicit architecture assumptions.
2024	LSH-MoE addressed communication overhead in large sparse-expert systems using locality-sensitive hashing. [10]	Reinforces the need to distinguish arithmetic sparsity from routing/ communication cost.
2024	Zhou and Liu developed a hybrid quantum-classical framework for estimating nonlinear functions of quantum states. [11]	Shows that quantum "nonlinearity" often requires copies, measurement, and classical post-processing rather than a simple nonlinear unitary.
2025	Barney, Lakhdar-Hamina, and Galitski proposed quantum neural activations implemented using gates and measurements, including ancilla-based weak-measurement constructions. [12]	Concrete evidence that activation-like quantum mechanisms require an explicit physical construction.
2025	Chakraborty et al. proposed QSVT methods avoiding traditional block encodings for certain Hamiltonian settings. [13]	Advances operator processing but does not turn arbitrary nonlinear classical maps into unitaries.

Year	Development	Relevance to HTNLA
2026	A recent Heterogeneous Graph Transformer preprint extends type-sensitive modeling into hyperbolic geometry. [14]	Shows that heterogeneous/type-specific representation learning remains an active architectural research area; it does not establish HTNLA performance.

Formal model and reconstruction semantics

Let the source label set be S and the target label set be T. Associate each source label s with dimension d_s and each target label t with dimension e_t:

$$\mathcal{H}_S = \bigoplus_{s \in S} \mathbb{R}^{d_s}, \qquad \mathcal{H}_T = \bigoplus_{t \in T} \mathbb{R}^{e_t}.$$

An element $H \in \mathcal{H}_S$ is therefore not required to have equally sized components:

$$H = (x_s)_{s \in S}, \qquad x_s \in \mathbb{R}^{d_s}.$$

This is the essential invariant inherited from HTLA: dimensional heterogeneity is explicit in the type of each block rather than hidden by assuming every label has a common vector size. The governing Nexus specification requires HTNLA to preserve those label boundaries. filecite turn0file1

Let $E \subseteq T \times S$ be the configured coupling relation. For each $(t, s) \in E$,

$$W_{t,s} \in \mathbb{R}^{e_t \times d_s}.$$

The target preactivation is

$$z_t = b_t + \sum_{s:(t,s)\in E} W_{t,s} x_s + r_t, \qquad b_t, r_t \in \mathbb{R}^{e_t},$$

followed by a target-specific map

$$y_t = \phi_t(z_t), \qquad \phi_t : \mathbb{R}^{e_t} \to \mathbb{R}^{e_t}.$$

Where configured,

$$y_t \leftarrow K_{k_t}(y_t),$$

where K_{k_t} denotes a deterministic k_t-winner selection rule. These are the canonical subsystem equations. filecite turn0file1

The multilayer construction is just composition of such maps:

$$H^{(k+1)} = \sigma^{(k)}\left(M^{(k)} H^{(k)} + b^{(k)}\right).$$

A skip/residual map is legal only when its block shapes make the sum well-typed:

$$H^{(k+1)} = R^{(k)} H^{(k)} + \sigma^{(k)} \left(M^{(k)} H^{(k)} + b^{(k)} \right).$$

Thus the residual connection is not “free”; for every contributing source-target pair, $R_{t,s}^{(k)}$ must have a shape compatible with the target component being added.

Why this is not a nonlinear algebra

Consider two maps

$$F_1(H) = \sigma(M_1 H), \qquad F_2(H) = \sigma(M_2 H).$$

In general,

$$F_1(H) + F_2(H) \;/\; \sigma((M_1 \nplus M_2)H).$$

The left side is a sum of two post-activation states, while the right side activates the sum of preactivations. Nonlinear σ does not distribute over addition in general. Therefore the parameterization is not closed under the ordinary operator-space addition that would justify calling it an algebra.

A cleaner mathematical object is a **network family**. One may also define a small category in which objects are labeled finite-dimensional direct sums and morphisms are finite compositions of dimensionally compatible HTNLA maps. Identity maps are represented by identity blocks and identity activation, and composition of two well-typed maps is again a well-typed nonlinear map. This is a structural definition, not a claim that arbitrary nonlinear maps form a linear algebra.

The historical phrase “graded nonlinear algebra” should consequently be classified `HISTORICAL/SUPERSEDED`. The useful graded/labeled structure remains in the underlying HTLA objects and block interfaces; it should not be promoted into an algebraic property that nonlinearity has destroyed.

Exact inversion and information loss

Suppose

$$F = \phi \circ A$$

for an affine precursor A. If there are two reachable states $z_1 \;/\; z_2$ ~~s~~such that

$$\phi(z_1) = \phi(z_2),$$

and $A(x_1) = z_1,\ A(x_2) = z_2$, then

$$F(x_1) = F(x_2)$$

despite $x_1 \;/\; x_2$. ~~H~~Hence F is not injective, and no single-valued exact inverse exists over that domain. This is a direct consequence of the definition of injectivity.

For common examples:

$$\mathrm{ReLU}(-1) = \mathrm{ReLU}(0) = 0,$$

so ReLU is non-injective.

By contrast, for $a > 0$,

$$\mathrm{LReLU}_a(x) = \begin{cases} x, & x \geq 0, \\ ax, & x < 0 \end{cases}$$

is bijective on $\mathbb{R}$, with inverse

$$\mathrm{LReLU}_a^{-1}(y) = \begin{cases} y, & y \geq 0, \\ y/a, & y < 0. \end{cases}$$

Likewise, $\tanh : \mathbb{R} \to (-1, 1)$ is bijective with inverse artanh. Therefore, the old argument that tanh automatically causes mathematical non-invertibility is incorrect. Numerical saturation, quantization, an unknown bias, a non-injective preceding block, dimensional compression, or subsequent sparsification can destroy practical or exact recovery; tanh itself is not the decisive reason.

Most importantly, k-WTA with $k < n$ is non-injective. Fix k coordinates with magnitudes strictly above a threshold. Any changes to the remaining $n - k$ coordinates that stay below the smallest winning magnitude leave the k-WTA output unchanged. Infinitely many preimages can therefore map to one sparse output.

This is precisely why invertible neural architectures such as RealNVP explicitly construct bijective transformations instead of expecting generic neural layers to remain invertible, while SurVAE explicitly models transformations for which reverse inference is not unique. [2]

Reconstruction methods versus quinary result states

The Nexus five-way taxonomy is best interpreted as **how reconstruction is attempted and what guarantee is available**, whereas the quinary four-state taxonomy is the **status returned to the caller**. The latter must remain authoritative. filecite turn0file1

HTNLA reconstruction class	Mathematical interpretation	Permissible quinary outcome
Exact inverse	A unique preimage is established under the relevant domain assumptions.	`EXACT`; `IMPOSSIBLE` if the requested output is proven outside the map's range.

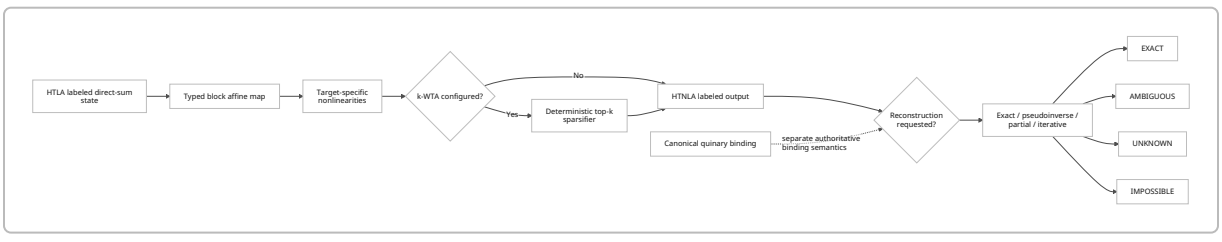
HTNLA reconstruction class	Mathematical interpretation	Permissible quinary outcome
Pseudoinverse	A least-squares or minimum-norm representative is selected; it need not be the originating state.	Usually `AMBIGUOUS`; `EXACT` only with independent proof of uniqueness; `IMPOSSIBLE` if exact consistency is required and disproven.
Partial unbinding	Some components or constraints are recoverable while others are not.	`AMBIGUOUS` if multiple completions remain; `UNKNOWN` where unresolved information cannot be inferred; `EXACT` only if all semantically required degrees of freedom are uniquely fixed.
Iterative/ resonator reconstruction	Recovery is a search or optimization problem rather than an analytic inverse.	Any of the four states, depending on certification: unique solution → `EXACT`; multiple viable solutions → `AMBIGUOUS`; unresolved search → `UNKNOWN`; proved absence of a valid preimage → `IMPOSSIBLE`.
Explicit unknown/ ambiguous result	Information loss or insufficient evidence is represented directly.	`UNKNOWN` or `AMBIGUOUS`; never silently promoted to `EXACT`.

A small but important consequence follows for pseudoinverses. For a linear A, the Moore–Penrose solution

$$\hat{x} = A^{+}y$$

may be the least-squares or minimum-norm solution, but unless A is injective on the relevant domain and the observations are consistent, $\hat{x}$ is not proof that the original x has been recovered. QSVT work even includes quantum procedures related to matrix pseudoinverses, but this remains linear-operator inversion, not recovery of information destroyed by arbitrary nonlinear many-to-one mappings. [15]

The deterministic quinary binding operation remains the canonical binding reference and should not be replaced by a circular-convolution-plus-tanh rule. A differentiable surrogate may be introduced for an application-specific training procedure only if it is explicitly `APPLICATION-SPECIFIC` and its outputs never override the deterministic reference semantics. The governing specification describes the quinary binding as intentionally information-losing and requires its `EXACT / AMBIGUOUS / UNKNOWN / IMPOSSIBLE` states to remain authoritative. 【filecite】turn0file1【】



Sparsity, expressivity, biological analogies, and quantum boundary

What k-WTA actually changes

For coupling graph E, the arithmetic cost of the dense block-affine part is

$$C_{\text{block}} = \Theta\left(\sum_{(t,s)\in E} e_t d_s\right).$$

This is the relevant expression; a blanket $O(D^2)$ statement hides the topology of the labels.

If every source component communicates densely with every target component and the total source and target dimensions are both proportional to D, the sum can indeed scale as $\Theta(D^2)$. But applying k-WTA **after** those products does not change the work already performed:

$$x \xrightarrow{\text{dense block product}} z \xrightarrow{\text{kWTA}} \tilde{z}.$$

A fixed retention fraction $\rho \in (0,1)$ can reduce the number of active values entering a later operation, provided the downstream implementation actually uses a sparse kernel. Because ρ is constant,

$$\rho D = \Theta(D),$$

so the change is a constant-factor change, not a new asymptotic class.

A genuine asymptotic improvement can arise from the coupling structure. Suppose there are L labels, each of bounded dimension n, and each target receives from at most c source labels for a constant c. Then

$$|E| = O(L),$$

and

$$C_{\text{block}} = O(Ln^2).$$

With total dimension $D = Ln$,

$$C_{\text{block}} = O(Dn),$$

which becomes linear in D when n is bounded. Here the improvement comes from **bounded-degree block connectivity**, not from k-WTA.

Similarly, a structured block that admits an FFT-like algorithm can achieve its own subquadratic cost, but that is a property of the block representation, not of sparsifying its output. The legacy derivation that a fixed sparsity percentage by itself yields $O(D \log D)$ is therefore `HISTORICAL/SUPERSEDED`.

The research literature supports a more modest statement. WTA mechanisms directly enforce sparse representations, and theoretical sparse-network work shows that carefully designed conditional subnetworks can retain approximation capability while activating fewer units. Neither result proves that an

arbitrary HTNLA layer automatically becomes numerically stable, preserves concentration of measure, or obtains an asymptotic improvement merely because a fixed fraction of its output is zero. [16]

Claim status: "k-WTA can be useful as a target-wise sparsification operator" is `PROPOSED` for HTNLA and well motivated by prior work. "k-WTA preserves HTNLA concentration-of-measure guarantees" remains `RESEARCH HYPOTHESIS`. A literature search identified specialized WTA representation and robustness results, but no general theorem sufficient to justify the latter claim for arbitrary learned HTNLA blocks. [17]

Expressivity without the universal-approximator shortcut

A sufficiently general HTNLA family can emulate an ordinary feed-forward network: select a set of labels, make the block coupling dense, and choose ordinary componentwise activations. Therefore some HTNLA configurations fall within model classes covered by standard approximation theorems.

But the theorem hypotheses matter. Cybenko proves density results for feed-forward networks with appropriate sigmoidal nonlinearities and sufficiently many units; the result is not a certificate for every finite architecture. [9]

Accordingly:

$$\boxed{\text{HTNLA family may contain universal approximators} \quad /}$$

Any stronger result for a specified block topology, fixed width, k-WTA level, or residual architecture requires its own theorem.

Biologically named learning rules

The safest core specification contains **no biologically named mechanism at all**. If a downstream application nevertheless wants a reward-modulated update, the Nexus three-level rule should be visible directly in the specification.

A generic computational construction could be

$$e_{ij}(t+1) = \lambda e_{ij}(t) + g(x_i(t), y_j(t))\,,$$

$$\Delta W_{ij}(t) = \eta\, m(t)\, e_{ij}(t),$$

where e_{ij} is an eligibility variable and $m(t)$ a scalar modulatory signal.

Its interpretation must be:

Level	Permissible statement
Level 1 — computational variable	$m(t)$ is a scalar software signal and e_{ij} is a stored eligibility trace. This is directly implementable and testable.

Level	Permissible statement
Level 2 — functional analogue	The arrangement is loosely analogous to reward-modulated plasticity in which delayed modulation acts on eligibility generated by earlier activity. Izhikevich and later reward-modulated-STDP analyses provide legitimate precedent for this functional pattern. [5]
Level 3 — phenomenological claim	No such claim follows. Implementing the variables does not establish a biological neurotransmitter mechanism, biological equivalence, or any phenomenological property.

Izhikevich's 2007 model links STDP-related eligibility to delayed dopamine modulation; Frémaux, Sprekeler, and Gerstner analyze functional requirements for reward-modulated STDP; Legenstein, Pecevski, and Maass provide an analytic learning treatment. These papers justify a **functional analogy**, but not the legacy HTNLA draft's particular sigmoid offsets, temperature constants, or risk formula. [5]

The old `risk_aware_modulation` prototype is not suitable for publication. As identified in the Nexus rewrite brief, its stated example attempts to add vectors of incompatible dimensions, and its “risk” term computes the variance of repeated copies of the same scalar reward, which is identically zero. filecite turn0file1 A replacement should use an actual history or modeled reward distribution if an application genuinely requires risk estimation, and it should be tagged `APPLICATION-SPECIFIC` or `INFERRED`, not represented as neuroscience.

Quantum boundary

The quantum section requires a sharper correction than a hedge.

A unitary map is linear on quantum state amplitudes. QSVT obtains extraordinary flexibility by applying **polynomial transformations to singular values of a linear operator encoded in or accessed through a quantum process**. That is not equivalent to asserting that an arbitrary nonlinear state map

$$F : \mathbb{C}^{d_1} \to \mathbb{C}^{d_2}$$

can simply be inserted as a block of a unitary U_F. [15]

Recent research illustrates the real difficulty. Zhou and Liu's 2024 treatment of nonlinear functions of quantum states employs a hybrid quantum-classical procedure involving coherent operations on copies of a state, randomized measurements, and classical post-processing. [11] Barney and colleagues' 2025 quantum neural-network proposal implements activation-like effects through quantum gates plus measurement, including ancilla-mediated weak-measurement mechanisms. [12] Even recent QSVT work that reduces or removes conventional block-encoding overhead remains an operator-polynomial framework rather than a license to realize arbitrary classical nonlinearity as deterministic unitary evolution. [13]

Therefore:

`RESEARCH HYPOTHESIS`: a future quantum-assisted HTNLA implementation might approximate or emulate selected nonlinear labeled maps through measurement-mediated, multi-copy, probabilistic, or hybrid quantum-classical procedures.

The previous generic claim that an arbitrary nonlinear F can be "block-encoded into a unitary" should be `HISTORICAL/SUPERSEDED`.

Reference implementation and executed validation

The following implementation intentionally does less than the legacy prototype. It provides only the part that can be specified rigorously: dimensionally typed labeled vectors, block-affine transformations, per-target nonlinearities, an outside-activation residual map matching the canonical multilayer form, optional k-WTA, and fail-fast dimension checks.

No training algorithm, biological rule, learned inverse, performance claim, or binding replacement is hidden inside it.

```
from __future__ import annotations

from dataclasses import dataclass
from typing import Callable, Mapping, Optional

import numpy as np

Array = np.ndarray
Activation = Callable[[Array], Array]


def identity(x: Array) -> Array:
    return x


def relu(x: Array) -> Array:
    return np.maximum(x, 0.0)


def leaky_relu(x: Array, slope: float = 0.1) -> Array:
    """Bijective componentwise map when slope > 0."""
    if not (0.0 < slope <= 1.0):
        raise ValueError("slope must lie in (0, 1].")
    return np.where(x >= 0.0, x, slope * x)


def k_wta(x: Array, k: int) -> Array:
    """
    Keep exactly k entries with largest absolute magnitude.
```

```
    Ties are resolved deterministically by lower coordinate index.
    """
    x = np.asarray(x, dtype=float)

    if x.ndim != 1:
        raise ValueError("k_wta expects a 1-D vector.")

    if not (1 <= k <= x.size):
        raise ValueError("k must satisfy 1 <= k <= len(x).")

    # Primary key: descending |x|.
    # Secondary key: ascending coordinate index.
    idx = np.lexsort((np.arange(x.size), -np.abs(x)))[:k]

    out = np.zeros_like(x)
    out[idx] = x[idx]
    return out


@dataclass(frozen=True)
class LabeledVector:
    components: Mapping[str, Array]

    def checked(self) -> "LabeledVector":
        checked: dict[str, Array] = {}

        for label, value in self.components.items():
            arr = np.asarray(value, dtype=float)

            if arr.ndim != 1:
                raise ValueError(
                    f"{label}: component must be a 1-D vector."
                )

            checked[label] = arr

        return LabeledVector(checked)


@dataclass
class NonlinearBlockOperator:
    """
    blocks[target][source] has shape:
        (target_dimension, source_dimension)
    """

    blocks: Mapping[str, Mapping[str, Array]]
```

```
    biases: Mapping[str, Array]
    activations: Mapping[str, Activation]

    # Optional R H term in:
    # H_next = R H + sigma(M H + b)
    residual_blocks: Optional[
        Mapping[str, Mapping[str, Array]]
    ] = None

    # Optional target-specific k in k-WTA.
    winners: Optional[Mapping[str, int]] = None

    def apply(self, x: LabeledVector) -> LabeledVector:
        x = x.checked()
        out: dict[str, Array] = {}

        for target, by_source in self.blocks.items():
            if (
                target not in self.biases
                or target not in self.activations
            ):
                raise KeyError(
                    f"{target}: missing bias or activation."
                )

            z = np.asarray(
                self.biases[target], dtype=float
            ).copy()

            if z.ndim != 1:
                raise ValueError(
                    f"{target}: bias must be 1-D."
                )

            target_dim = z.size

            # z_t = b_t + sum_s W_{t,s} x_s
            for source, W in by_source.items():
                if source not in x.components:
                    raise KeyError(
                        f"missing source label {source!r}"
                    )

                W = np.asarray(W, dtype=float)
                xs = x.components[source]

                expected = (target_dim, xs.size)
```

```
            if W.shape != expected:
                raise ValueError(
                    f"block {target}<-{source} "
                    f"has shape {W.shape}, "
                    f"expected {expected}"
                )

            z += W @ xs

        # sigma_t(z_t)
        y = np.asarray(
            self.activations[target](z),
            dtype=float,
        )

        if y.shape != (target_dim,):
            raise ValueError(
                f"{target}: activation changed "
                "the target-vector shape."
            )

        # Optional R H skip term, outside sigma.
        if (
            self.residual_blocks
            and target in self.residual_blocks
        ):
            r = np.zeros(target_dim, dtype=float)

            for source, R in (
                self.residual_blocks[target].items()
            ):
                if source not in x.components:
                    raise KeyError(
                        "missing residual source "
                        f"label {source!r}"
                    )

                R = np.asarray(R, dtype=float)
                xs = x.components[source]
                expected = (target_dim, xs.size)

                if R.shape != expected:
                    raise ValueError(
                        f"residual block "
                        f"{target}<-{source} "
                        f"has shape {R.shape}, "
                        f"expected {expected}"
                    )
```

```
                    r += R @ xs

                y = r + y

            if (
                self.winners
                and target in self.winners
            ):
                y = k_wta(
                    y,
                    self.winners[target],
                )

            out[target] = y

        return LabeledVector(out)
```

The exact implementation above was executed with genuinely heterogeneous source dimensions:

```
rng = np.random.default_rng(7)

x = LabeledVector(
    {
        "text": rng.normal(size=3),
        "vision": rng.normal(size=5),
    }
)

operator = NonlinearBlockOperator(
    blocks={
        "semantic": {
            "text": rng.normal(size=(4, 3)),
            "vision": rng.normal(size=(4, 5)),
        },
        "control": {
            "vision": rng.normal(size=(2, 5)),
        },
    },
    biases={
        "semantic": rng.normal(size=4),
        "control": rng.normal(size=2),
    },
    activations={
        "semantic": leaky_relu,
        "control": np.tanh,
```

```
    },
    residual_blocks={
        "semantic": {
            "text": rng.normal(size=(4, 3)),
        }
    },
    winners={
        "semantic": 2,
    },
)

y = operator.apply(x)

print(
    {
        label: vector.shape
        for label, vector in y.components.items()
    }
)

print(
    np.count_nonzero(
        y.components["semantic"]
    )
)
```

Executed output:

```
{'semantic': (4,), 'control': (2,)}
2
```

Thus a three-dimensional source label and a five-dimensional source label were simultaneously mapped into independently sized four-dimensional and two-dimensional target labels, and the target configured for k-WTA retained the requested two entries.

A negative shape test was also executed:

```
bad = NonlinearBlockOperator(
    blocks={
        "semantic": {
            # Wrong: text has dimension 3, not 4.
            "text": np.zeros((4, 4))
        }
    },
    biases={
```

```
        "semantic": np.zeros(4)
    },
    activations={
        "semantic": identity
    },
)

bad.apply(x)
```

The operator rejected the malformed block:

```
ValueError:
block semantic<-text has shape (4, 4),
expected (4, 3)
```

This validation is small, but it verifies an important subsystem invariant that the older `dynamic_label_adapt` prototype violated: **different labels are allowed to have genuinely different dimensions, and cross-label computation must occur through an explicitly dimensioned map rather than by blindly stacking and averaging heterogeneous arrays.** The earlier defect is documented in the rewrite specification. filecite turn0file1

The code should still be classified modestly. The dimensional contract and demonstrated fixture are `VERIFIED`; the class is a `PROPOSED` reference implementation for the subsystem. It has not been demonstrated as a production database integration, distributed training system, high-performance sparse kernel, or benchmark-winning retrieval engine.

Evaluation, stakeholders, risks, and strategic recommendations

A rigorous HTNLA evaluation should begin with **questions**, not with preloaded target numbers.

The primary empirical hypothesis is:

$$H_1: \quad \text{for tasks whose relevant decision or retrieval geometry is nonlinear, a trained HTNLA configuration can outperform a matched HTLA baseline under at least one preregistered accuracy-efficiency tradeoff.}$$

This is a `RESEARCH HYPOTHESIS`. It should remain unquantified until experiments are run.

A credible benchmark should compare at least these conditions:

Variant	Purpose
Linear HTLA baseline	Establish what the existing labeled block system already accomplishes.
HTNLA without k-WTA	Measure the effect of nonlinearity itself.
HTNLA with k-WTA	Separate sparsification effects from nonlinear transformation effects.
Structurally block-sparse HTNLA	Test whether sparse label connectivity yields actual compute savings.
Invertible-profile HTNLA	Measure the cost and benefit of restricting blocks/activations to preserve recoverability.
Parameter- or compute-matched conventional network	Determine whether any benefit comes from HTNLA's labeled structure rather than merely more trainable capacity.

Evaluation should use both synthetic and application-level data. **Synthetic heterogeneous tests** should deliberately vary component dimensions and include functions with known unique, multiple, and nonexistent preimages so that `EXACT / AMBIGUOUS / UNKNOWN / IMPOSSIBLE` behavior can be verified. **Application-specific retrieval data** should contain the heterogeneous modalities or feature families the actual system is intended to serve. No dataset should be called an “HTNLA benchmark” merely because it was used after the architecture was designed; task choice, splits, baselines, and stopping criteria should be preregistered where claims will be consequential.

The metrics should be ordinary, reproducible engineering metrics rather than undefined proxies:

Dimension	Recommended measurements	What would count as evidence
Retrieval quality	Recall@k, MRR, NDCG or task-appropriate ranking metric	Paired comparison with baseline, uncertainty intervals and seeds
Predictive performance	Task loss, accuracy/F-score where applicable	Held-out evaluation
Reconstruction	Forward residual $\|F(\hat{x}) - y\|$, unique-preimage checks where possible	Correct quinary status plus reconstruction error
Epistemic behavior	Frequency of `EXACT`, `AMBIGUOUS`, `UNKNOWN`, `IMPOSSIBLE`	Calibration against problems with known ground truth
Sparsity	Fraction of active coordinates, active blocks, actual sparse-kernel utilization	Distinguishes logical zeros from realized compute savings
Efficiency	End-to-end latency percentiles, throughput, memory footprint	Measured on specified hardware and workload
Energy	Joules/query or joules/training step from hardware telemetry	Direct measurement, not a fixed “conservation” percentage

Dimension	Recommended measurements	What would count as evidence
Training stability	Loss trajectories, gradient norms, failed-run rate	Seeded repeated runs
Index quality, if ANN is used	Recall/latency after transforming both indexed items and queries consistently	Comparison against the same retrieval budget

No fixed threshold is justified before baselines exist. In particular, arbitrary values for an undefined “binding score,” a percentage change in an undefined consciousness-related quantity, or a fixed energy-law percentage provide no scientifically useful validation of a nonlinear block-map subsystem. The governing rewrite specification correctly requires those kinds of claims to be omitted unless given an explicit Level-1 computational definition and actual measurement. filecite turn0file1

Stakeholder model

Stakeholder	Decision responsibility
Nexus specification maintainers	Ratify canonical equations, label semantics, reconstruction statuses, and claim-status rules.
HTLA/HTNLA runtime engineers	Enforce dimensional contracts, residual semantics, serialization/ versioning, deterministic k-WTA behavior, and failure handling.
ML/retrieval engineers	Train nonlinear blocks, maintain matched baselines, evaluate transformed-index consistency, and run ablations.
Reliability and verification owners	Test inverse-status correctness, malformed configurations, reproducibility, numerical failure modes, and abstention behavior.
Application owners	Decide whether the added nonlinear capacity improves a real task enough to justify operational complexity.
Research reviewers	Determine whether proposed biological, quantum, or approximation claims remain within the evidence actually provided.

Principal risks and uncertainties

The highest mathematical risk is **silent information loss**. ReLU, dimensional compression, k-WTA, rank-deficient blocks, quantization, or ordinary data collisions can make the map many-to-one. The mitigation is not to promise a better optimizer; it is to make the reconstruction result type part of the public contract.

The highest architectural risk is **geometry drift between learned transformation and retrieval index**. If stored vectors are indexed under one representation but queries are transformed under a changed nonlinear map, nearest-neighbor relationships can cease to mean what the index assumes. A production implementation therefore needs model-version coupling between stored representations, query transforms, and indexes, with rebuild or migration procedures whenever the learned map changes materially. This is an engineering consequence of indexing transformed representations, not evidence that nonlinear preprocessing automatically produces “manifold-aware sharding.”

The highest performance risk is **sparsity theater**: an activation tensor may contain many zeros while the system still executes dense matrix kernels, dense memory transfers, dense network communication, or an expensive top-k selection. Recent sparse-network and MoE work is useful precisely because it analyzes or engineers the routing and active-computation path explicitly. [8]

The highest research-communication risk is **category inflation**: calling an ordinary computational modulator "dopamine," calling an engineered integration statistic a property of experience, or labeling a target percentage a physical conservation law. Reward-modulated STDP literature makes a functional analogy possible, but it does not license arbitrary neuroscience terminology for unrelated formulas. [18]

The highest long-horizon risk is **quantum overclaiming**. Linear-operator processing such as QSVT is a mature theoretical framework, but arbitrary nonlinear state transformations require additional machinery and should not be implied by the word "block encoding." [19]

Strategic recommendations

Priority	Recommendation	Required evidence before promotion
Core specification	Define HTNLA as a family of labeled nonlinear maps over HTLA, not a "graded nonlinear algebra."	Mathematical type-checking and ratification against the Nexus equations.
Reconstruction contract	Make the five reconstruction classes and four quinary result states first-class API concepts.	Test suite with known unique, multiple, absent, and underdetermined preimages.
Binding discipline	Keep deterministic quinary binding authoritative; do not revive tanh-convolution binding as a competing canonical operator.	Any training surrogate must be separately specified and tagged `APPLICATION-SPECIFIC`.
Performance claims	Delete fixed recall-gain claims until matched ablations exist.	Repeated held-out experiments with baselines, seeds, uncertainty estimates, and workload specification.
Sparsity	Treat k-WTA as an optional representation operator; pursue actual compute reduction through block sparsity or sparse-aware kernels.	Hardware-level latency, utilization, memory and energy measurements.
Invertible profile	Define an optional subset using dimension-compatible blocks and injective/invertible components.	Forward/reverse property tests and numerical conditioning analysis.
Biological analogies	Keep any reward-modulated mechanism outside the core and explicitly Level-1/ Level-2 only.	Exact equation, implementation, and literature mapping; no Level-3 implication.

Priority	Recommendation	Required evidence before promotion
Quantum work	Move quantum HTNLA to a clearly separate `RESEARCH HYPOTHESIS` track.	Concrete circuit/resource model demonstrating how each nonlinear effect is produced.
Governance	Require a claim-status tag whenever a statement moves beyond definitions or directly reproduced measurements.	Source, derivation, or executable artifact attached to the claim.
Roadmap	Replace month-by-month promises and arbitrary latency targets with stage gates.	Advancement only after unit tests → synthetic validation → matched benchmark → production telemetry.

The immediate development order should therefore be **specification → reference implementation → reconstruction tests → controlled ablations → sparse-performance work → index/runtime integration**. There is no evidentiary reason to begin with more elaborate claims.

Claim-status audit and annotated bibliography

The Nexus vocabulary supplied in the rewrite brief is retained unchanged: `CANONICAL, IMPLEMENTED, VERIFIED, PARTIALLY IMPLEMENTED, ACTIVE DEVELOPMENT, PROPOSED, DESIGN TARGET, EXPERIMENTAL, RESEARCH HYPOTHESIS, APPLICATION-SPECIFIC, HISTORICAL/SUPERSEDED, INFERRED`.
filecite
turn0file1

Closing claim-status audit

Claim or artifact	Status	Basis
Canonical block/nonlinear layer equations	`CANONICAL`	Given explicitly by the Nexus v4 controlling specification.
filecite
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Five reconstruction classes	`CANONICAL`	Nexus v4 governing specification.
filecite
turn0file1
Quinary `EXACT / AMBIGUOUS / UNKNOWN / IMPOSSIBLE` output contract	`CANONICAL`	Nexus v4 governing specification.
filecite
turn0file1
Calling the nonlinear-map family a “graded nonlinear algebra”	`HISTORICAL/ SUPERSEDED`	Fails closure/linearity requirements under ordinary operator addition; corrected by direct derivation.
Treating HTNLA as composable labeled nonlinear maps	`PROPOSED`	Mathematically coherent reframing elaborated here.

Claim or artifact	Status	Basis
Reference `NonlinearBlockOperator` shown above	`PROPOSED`	Suitable minimal reference design, not claimed production implementation.
Heterogeneous-dimensional execution of that reference operator	`VERIFIED`	Executed with distinct source/target dimensions and malformed-shape negative test in this research run.
k-WTA produces a sparse target representation	`VERIFIED` as an operator property	By implementation/definition; prior WTA literature independently demonstrates explicit sparsification use. [20]
k-WTA generically preserves concentration of measure after arbitrary learned nonlinear maps	`RESEARCH HYPOTHESIS`	No adequate general theorem was identified; available results are architecture-specific. [17]
Fixed fractional k-WTA changes dense $O(D^2)$ multiplication to $O(D \log D)$	`HISTORICAL/ SUPERSEDED`	Contradicted by direct operation-count analysis.
Structural block sparsity can improve asymptotic compute scaling	`VERIFIED` mathematically for the stated coupling assumptions	Follows from $\Theta(\sum_{(t,s)\in E} e_t d_s)$.
Predetermined percentage improvement in associative recall	`HISTORICAL/ SUPERSEDED`	No experiment in the source establishes it.
Nonlinear HTNLA can improve tasks whose relevant geometry is nonlinear	`RESEARCH HYPOTHESIS`	Plausible motivation requiring matched empirical testing.
Every fixed HTNLA configuration is a universal approximator	Not established; previous blanket form `HISTORICAL/ SUPERSEDED`	Universal-approximation theorems require architectural/activation/ capacity hypotheses. [9]
The HTNLA family can contain network subclasses covered by ordinary approximation theorems	`INFERRED`	Follows when its block configuration reduces to a theorem-covered feed-forward architecture. [9]
Generic reward-modulated eligibility rule as a software mechanism	`APPLICATION-SPECIFIC`	Legitimate computational mechanism; only a Level-2 functional analogy to reward-modulated STDP literature. [5]
Legacy specific neuromodulator equations/constants as biologically grounded	`HISTORICAL/ SUPERSEDED`	External literature supports the general modulation concept, not those particular formulas.

Claim or artifact	Status	Basis
Fixed “energy conservation” percentage as a measured property	`HISTORICAL/SUPERSEDED`	Nexus canon identifies it as a design motif; real energy claims require telemetry. filecite turn0file1
Arbitrary nonlinear F can routinely be block-encoded into a unitary	`HISTORICAL/SUPERSEDED`	QSVT concerns polynomial transformations of encoded linear operators; nonlinear quantum tasks need additional mechanisms. [19]
Quantum-assisted HTNLA using measurement/hybrid nonlinear mechanisms	`RESEARCH HYPOTHESIS`	Plausible direction supported by recent adjacent research, not an implemented subsystem. [21]
Narrow novelty claim based on labels + nonlinear maps + explicit epistemic reconstruction contract	`INFERRED`	Prior art contains the individual ingredients; the specified combination is the defensible differentiation identified by this review. [22]

Internal corpus sources. The legacy HTNLA blueprint is useful primarily as the historical design source whose equations, prototypes, claims, and architectural ideas are being normalized. filecite turn0file0 The Nexus v4 rewrite specification is the controlling internal source for canonical equations, binding compatibility, reconstruction status semantics, epistemic vocabulary, and the required corrections. filecite turn0file1

Annotated external bibliography — top sources

Schlichtkrull, Kipf, Bloem, van den Berg, Titov & Welling — “Modeling Relational Data with Graph Convolutional Networks” (2017/2018). R-GCN is essential novelty-scoping prior art because it makes transformations relation-specific in a multi-relational representation system. It demonstrates that typed transformation blocks are an established neural architecture pattern; HTNLA must distinguish itself through its labeled transdimensional contract and reconstruction semantics rather than by claiming type-specific blocks as novel. [23]

Hu, Dong, Wang & Sun — “Heterogeneous Graph Transformer” (2020). HGT uses node- and edge-type-dependent parameters and heterogeneous attention to maintain representations across multiple types. It is one of the closest architectural comparisons for per-label/per-type learned transformations and reinforces the need for a conservative HTNLA novelty claim. [24]

Baykal, Dikkala, Panigrahy, Rashtchian & Wang — “A Theoretical View on Sparsely Activated Networks” (2022). This work is valuable because it replaces vague claims about sparsity with an explicit formal model of data-dependent sparse networks and derives approximation results for a particular routing construction. It supports exploring structurally sparse HTNLA execution while underscoring that sparsity benefits must be proven under a stated computation model. [3]

Dinh, Sohl-Dickstein & Bengio — "Density Estimation Using Real NVP" (2016). RealNVP is a foundational example of designing learnable transformations so that invertibility and exact reverse computation are guaranteed by construction. It provides a strong contrast to generic HTNLA nonlinear layers, where invertibility cannot be assumed. [25]

Nielsen, Jaini, Hoogeboom, Winther & Welling — "SurVAE Flows: Surjections to Bridge the Gap between VAEs and Flows" (2020). SurVAE is especially useful for the Nexus reconstruction taxonomy because it explicitly reasons about bijective and surjective transformations rather than forcing every map into an invertible mold. It gives external precedent for treating reverse recovery as qualitatively different when information has been collapsed. [26]

Makhzani & Frey — "Winner-Take-All Autoencoders" (2015). This paper demonstrates WTA as a direct sparsification mechanism in representation learning. It supports the limited claim that WTA can enforce sparsity, but does not support a universal HTNLA concentration-of-measure or asymptotic-complexity theorem. [20]

Dasgupta & Tosh — "Expressivity of Expand-and-Sparsify Representations" (2020). The paper analyzes a random expansion followed by winner-take-all sparsification and proves approximation properties under specific assumptions. Its finding that ordinary WTA does not automatically obtain the desired manifold adaptivity is particularly useful as a warning against upgrading architectural intuition into a generic theorem. [17]

Frémaux, Sprekeler & Gerstner — "Functional Requirements for Reward-Modulated Spike Timing-Dependent Plasticity" (2010). This work provides appropriate scientific grounding for a narrowly stated Level-2 functional analogy involving reward modulation and local plasticity. It does not validate arbitrary formulas merely because their variables are given neurotransmitter names. [27]

Cybenko — "Approximation by Superpositions of a Sigmoidal Function" (1989). The classical universal-approximation result is the correct antidote to the legacy comparison table's unconditional "Universal Approximator" label. It shows that approximation theorems have explicit activation, capacity, domain, and architecture conditions; those conditions must be checked before transferring the result to an HTNLA configuration. [9]

Gilyén, Su, Low & Wiebe — "Quantum Singular Value Transformation and Beyond" (2018/2019). QSVT is the appropriate reference for what block-encoded quantum operator processing really establishes: polynomial transformations of singular values associated with linear operators. It sharply delineates that capability from the unsupported assertion that any arbitrary nonlinear classical map can simply become a unitary block. [15]

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[1] [22] [23] https://arxiv.org/abs/1703.06103
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[2] [25] https://arxiv.org/abs/1605.08803
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[3] [7] [8] https://arxiv.org/abs/2208.04461
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